

A swap partition can be created either as a primary or logical partition inside an extended partition, depending on partition availability.

Practical: Swap Partition Configuration

Question

Can use any available disk (like /dev/sda, /dev/sdb) instead of /dev/nvme0n1, depending on your system's device naming.

You are given a disk /dev/sdb with free space. Perform the following tasks:

Partitioning

- Create a primary partition of size 2GB on /dev/sdb
- Change its type to Linux Swap

Swap Configuration

- Initialize the partition as swap space
- Activate the swap partition
- Verify that swap is active

Temporary vs Permanent

Temporary:

- Activate swap immediately

Permanent:

- Configure swap in /etc/fstab using UUID

Verification

- Verify swap usage

Advanced Tasks (Not necessary)

- Set swap priority manually
- Create an additional swap file (1GB) and activate it
- Verify both swap partition and swap file are active

Cleanup (WITHOUT reboot)

- Disable swap
- Remove swap entry from /etc/fstab
- Disable and delete swap file
- Delete partition
- Apply changes

```
-----commands-----

# =====
# STEP 1: Verify disk
# =====

lsblk

# =====
# STEP 2: Create swap partition (2GB)
# =====

fdisk /dev/sdb
# n → p → 1 → Enter → +2G
# t → 82 (Linux swap)
# w

# Apply changes
partprobe

# Verify
lsblk

# =====
# STEP 3: Initialize + Activate Swap
# =====

mkswap /dev/sdb1
swapon /dev/sdb1

# Verify
swapon --show
free -h

# =====
# STEP 4: Permanent Configuration
# =====
```

```
blkid
```

```
vi /etc/fstab
```

```
# Add:
```

```
# UUID=xxxx none swap defaults 0 0
```

```
systemctl daemon-reload
```

```
swapon -a
```

```
# Verify
```

```
swapon --show
```

```
free -h
```

```
cat /proc/swaps
```

```
# =====
```

```
# STEP 5: (Optional) Advanced Tasks
```

```
# =====
```

```
# Set priority
```

```
swapon -p 10 /dev/sdb1
```

```
# Create swap file (1GB)
```

```
fallocate -l 1G /swapfile
```

```
chmod 600 /swapfile
```

```
mkswap /swapfile
```

```
swapon /swapfile
```

```
# Verify both
```

```
swapon --show
```

```
free -h
```

```
# =====
```

```
# STEP 6: Cleanup (WITHOUT reboot)
```

```
# =====
```

```
# Disable swap partition
```

```
swapoff /dev/sdb1
```

```
# Remove fstab entry
```

```
vi /etc/fstab
```

```
# Disable and remove swap file
```

```
swapoff /swapfile
```

```
rm /swapfile
```

```
# Delete partition
```

```
fdisk /dev/sdb
```

```
# d → 1 → w
```

```
# Apply changes
```

```
partprobe
```

```
# Final verify
```

```
Lsblk
```

```
-----x-----
```

Partitioning

- Create a primary partition of size 2GB on /dev/sdb

```
22:01:54 | root@localhost | ~ | cmd:178 | cpu: 0.00 | mem:27% | disk:21% | ip:
🚀 fdisk /dev/sda
```

Welcome to fdisk (util-linux 2.40.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): p

Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors

Disk model: VMware Virtual S

Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0x17c80010

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sda1		2048	4196351	4194304	2G	83	Linux

Command (m for help): n

Partition type

p primary (1 primary, 0 extended, 3 free)

e extended (container for logical partitions)

Select (default p): p

Partition number (2-4, default 2): 2

First sector (4196352-20971519, default 4196352):

Last sector, +/-sectors or +/-size{K,M,G,T,P} (4196352-20971519, default 20971519): +2G

Created a new partition 2 of type 'Linux' and of size 2 GiB.

Command (m for help): p

Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors

Disk model: VMware Virtual S

Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0x17c80010

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sda1		2048	4196351	4194304	2G	83	Linux
/dev/sda2		4196352	8390655	4194304	2G	83	Linux

- Change its type to Linux Swap

```

Command (m for help): t
Partition number (1,2, default 2): 2
Hex code or alias (type L to list all): l

00 Empty                27 Hidden NTFS Win  82 Linux swap / So  c1 DRDOS/sec (FAT-
01 FAT12                39 Plan 9           83 Linux            c4 DRDOS/sec (FAT-
02 XENIX root           3c PartitionMagic  84 OS/2 hidden or  c6 DRDOS/sec (FAT-
03 XENIX usr            40 Venix 80286      85 Linux extended  c7 Syrix
04 FAT16 <32M           41 PPC PReP Boot   86 NTFS volume set da Non-FS data
05 Extended             42 SFS             87 NTFS volume set db CP/M / CTOS / .
06 FAT16                4d QNX4.x          88 Linux plaintext de Dell Utility
07 HPFS/NTFS/exFAT      4e QNX4.x 2nd part 8e Linux LVM       df BootIt
08 AIX                  4f QNX4.x 3rd part 93 Amoeba         e1 DOS access
09 AIX bootable         50 OnTrack DM      94 Amoeba BBT      e3 DOS R/O
0a OS/2 Boot Manag     51 OnTrack DM6 Aux 9f BSD/OS         e4 SpeedStor
0b W95 FAT32            52 CP/M            a0 IBM Thinkpad hi ea Linux extended
0c W95 FAT32 (LBA)      53 OnTrack DM6 Aux a5 FreeBSD        eb BeOS fs
0e W95 FAT16 (LBA)      54 OnTrackDM6      a6 OpenBSD        ee GPT
0f W95 Ext'd (LBA)      55 EZ-Drive        a7 NeXTSTEP       ef EFI (FAT-12/16/
10 OPUS                 56 Golden Bow     a8 Darwin UFS     f0 Linux/PA-RISC b
11 Hidden FAT12         5c Priam Edisk     a9 NetBSD         f1 SpeedStor
12 Compaq diagnost     61 SpeedStor       ab Darwin boot    f4 SpeedStor
14 Hidden FAT16 <3      63 GNU HURD or Sys af HFS / HFS+     f2 DOS secondary
16 Hidden FAT16         64 Novell Netware b7 BSDI fs        f8 EBBR protective
17 Hidden HPFS/NTF      65 Novell Netware b8 BSDI swap      fb VMware VMFS
18 AST SmartSleep       70 DiskSecure Mult bb Boot Wizard hid fc VMware VMKCORE
1b Hidden W95 FAT3      75 PC/IX           bc Acronis FAT32 L fd Linux raid auto
1c Hidden W95 FAT3      80 Old Minix       be Solaris boot   fe LANstep
1e Hidden W95 FAT1      81 Minix / old Lin bf Solaris        ff BBT
24 NEC DOS

Aliases:
  linux      - 83
  swap       - 82
  extended   - 05
  uefi       - EF
  raid       - FD
  lvm        - 8E
  linuxex    - 85
Hex code or alias (type L to list all): 82

```

```

Hex code or alias (type L to list all): 82

Changed type of partition 'Linux' to 'Linux swap / Solaris'.

Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device      Boot      Start        End Sectors  Size Id Type
/dev/sda1                2048 4196351 4194304    2G 83 Linux
/dev/sda2             4196352 8390655 4194304    2G 82 Linux swap / Solaris

Command (m for help): w
The partition table has been altered.
Syncing disks.

```

```
22:03:23 | root@localhost | ~ | cmd:179 | cpu: 0.03 | mem:27% | disk:21% | ip: |  
🚀 partprobe
```

Swap Configuration

```
mkswap → swapon → verify → fstab → swapon -a → swapoff → delete
```

- Initialise the partition as swap space

```
22:07:56 | root@localhost | ~ | cmd:182 | cpu: 0.32 | mem:27% | disk:21% | ip:  
| git: | ssh: | time:0s  
🚀 mkswap /dev/sda2  
mkswap: /dev/sda2: warning: don't erase bootbits sectors  
      (dos partition table detected). Use -f to force.  
Setting up swspace version 1, size = 2 GiB (2147479552 bytes)  
no label, UUID=fa509a05-fd1b-4dbb-9182-65bbc30355b7
```

- Activate the swap partition

```
22:12:41 | root@localhost | ~ | cmd:183 | cpu: 0.04 | mem:27%  
| git: | ssh: | time:0s  
🚀 swapon /dev/sda2
```

- Verify that swap is active

```
22:13:23 | root@localhost | ~ | cmd:184 | cpu: 0.18 | mem:27% | disk:21% | ip: | git: | ssh: |  
🚀 swapon --show  
NAME      TYPE      SIZE USED PRIO  
/dev/dm-1 partition 2.5G  0B   -2  
/dev/sda2 partition  2G   0B   -3  
  
22:13:34 | root@localhost | ~ | cmd:185 | cpu: 0.15 | mem:27% | disk:21% | ip: | git: | ssh: |  
🚀 free -h  


|       | total | used  | free  | shared | buff/cache | available |
|-------|-------|-------|-------|--------|------------|-----------|
| Mem:  | 5.6Gi | 1.5Gi | 3.6Gi | 18Mi   | 662Mi      | 4.0Gi     |
| Swap: | 4.5Gi | 0B    | 4.5Gi |        |            |           |


```

Temporary vs Permanent

Temporary:

- Activate swap immediately

```
22:12:41 | root@localhost | ~ | cmd:183 | cpu: 0.04 | mem:27%  
| git: | ssh: | time:0s  
🚀 swapon /dev/sda2
```

Permanent:

- Configure swap in `/etc/fstab` using UUID

```
22:13:51 | root@localhost | ~ | cmd:186 | cpu: 0.12 | mem:27% | disk:21% | ip: | gtt: | ssh: | time:0s
❏ blkid
/dev/mapper/cs-root: UUID="e79aa976-2f04-4e51-8942-08170334e649" BLOCK_SIZE="512" TYPE="xfs"
/dev/sdb3: UUID="JvsA7G-0t1q-x02o-1iGb-yedU-K6uc-WVRD31" TYPE="LVM2_member" PARTUUID="2b5c3d62-67af-4337-a7b7-7f8d918ea142"
/dev/mapper/cs-swap: UUID="de55cc71-342c-4646-817d-a66a99e96439" TYPE="swap"
/dev/sdb2: UUID="b4c18b19-dfe8-4088-8806-ec8fa03b629a" BLOCK_SIZE="512" TYPE="xfs" PARTUUID="16b9aa22-0b3a-4829-b097-3d46938ee14b"
/dev/sda1: UUID="a1b98820-f4c5-43ad-ab2c-b5585b83d06f" BLOCK_SIZE="512" TYPE="xfs" PARTUUID="17c80010-01"
/dev/sdb1: PARTUUID="443939da-3edc-452b-a7c8-0c04c6dd17"
/dev/sda2: UUID="fa509a05-fd1b-4dcb-9182-65bbcb3035b7" TYPE="swap" PTTYPE="dos" PARTUUID="17c80010-02"

22:17:36 | root@localhost | ~ | cmd:187 | cpu: 0.00 | mem:27% | disk:21% | ip: | gtt: | ssh: | time:0s
vim /etc/fstab
```

```
# /etc/fstab
# Created by anaconda on Sun Mar 15 19:41:01 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=e79aa976-2f04-4e51-8942-08170334e649 / xfs defaults 0 0
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot xfs defaults 0 0
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none swap defaults 0 0

/dev/sda1 /mnt/sda1 xfs defaults 0 0
#-----

#/dev/sda2: UUID="fa509a05-fd1b-4dbb-9182-65bbc30355b7" TYPE="swap" PTTYPE="dos" PARTUUID="17c80010-02"
#/dev/sda2 none swap default 0 0

UUID=fa509a05-fd1b-4dbb-9182-65bbc30355b7 none swap default 0 0

~
~
~
~
~
~
~
~
```

:wq

```
22:22:31 | root@localhost | ~ | cmd:190 | cpu: 0.18 | mem:28% | d
| git: | ssh: | time:0s
🚀 systemctl daemon-reload

22:23:39 | root@localhost | ~ | cmd:191 | cpu: 0.06 | mem:28% | d
| git: | ssh: | time:0s
🚀 swapon -a

22:23:41 | root@localhost | ~ | cmd:192 | cpu: 0.06 | mem:28% | d
```


Verification

- Verify swap usage

```
22:23:41 | root@localhost | ~ | cmd:192 | cpu: 0.06 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 swapon --show
NAME      TYPE      SIZE USED  PRIO
/dev/dm-1 partition 2.5G  0B   -2
/dev/sda2 partition  2G   0B   -3

22:25:12 | root@localhost | ~ | cmd:193 | cpu: 0.01 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 free -h
              total        used        free      shared  buff/cache   available
Mem:           5.6Gi        1.6Gi        3.6Gi         18Mi        664Mi       4.0Gi
Swap:          4.5Gi          0B        4.5Gi

22:25:26 | root@localhost | ~ | cmd:194 | cpu: 0.01 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 cat /proc/swaps
Filename                                Type              Size              Used Priority
/dev/dm-1                               partition         2621436           0      -2
/dev/sda2                               partition         2097148           0      -3
```

Advanced Tasks|(not taught)

- Set swap priority manually
- Create an additional swap file (1GB) and activate it
- Verify both swap partition and swap file are active

```
22:28:58 | root@localhost | ~ | cmd:196 | cpu: 0.01 | mem:
| git: | ssh: | time:0s
🚀 fallocate -l 1G /swapfile

22:29:08 | root@localhost | ~ | cmd:197 | cpu: 0.07 | mem:
| git: | ssh: | time:0s
🚀 chmod 600 /swapfile

22:29:27 | root@localhost | ~ | cmd:198 | cpu: 0.06 | mem:
| git: | ssh: | time:0s
🚀 swapon /swapfile
```

Cleanup (WITHOUT reboot)

- Disable swap

```
22:34:00 | root@localhost | ~ | cmd:199 | cpu: 0.00 | mem:28% |
| git: | ssh: | time:0s
🚀 swapoff /dev/sda2
```

- Remove swap entry from /etc/fstab

```
# /etc/fstab
# Created by anaconda on Sun Mar 15 19:41:01 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=e79aa976-2f04-4e51-8942-08170334e649 /                xfs      defa
ultra    0 0
UUID=b41c8b19-dfe8-4088-8806-ec8fa03b629a /boot          xfs      defa
ultra    0 0
UUID=de55cc71-342c-4646-817d-ae6a99e96439 none           swap     defa
ultra    0 0

/dev/sda1      /mnt/sda1      xfs      defaults      0      0
#-----
---
|
~
~
~
~
~
~
~
```

- Disable and delete the swap file

```
22:37:05 | root@localhost | ~ |
| git: | ssh: | time:0s
🚀 swapoff /dev/sda2
```

```
22:37:47 | root@localhost | ~ |
| git: | ssh: | time:0s
🚀 rm /swapfile
```

- Delete partition

```
22:38:28 | root@localhost | ~ | cmd:206 | cpu: 0.08 | mem:28% | disk:21% | ip: | git: | ssh: | time:0s
🚀 fdisk /dev/sda

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

This disk is currently in use - repartitioning is probably a bad idea.
It's recommended to umount all file systems, and swapoff all swap
partitions on this disk.

Command (m for help): p

Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device      Boot  Start      End Sectors Size Id Type
/dev/sda1                2048 4196351 4194304  2G 83 Linux
/dev/sda2          4196352 8390655 4194304  2G 82 Linux swap / Solaris

Command (m for help): d
Partition number (1,2, default 2): 2

Partition 2 has been deleted.
```

```
Command (m for help): p
Disk /dev/sda: 10 GiB, 10737418240 bytes, 20971520 sectors
Disk model: VMware Virtual S
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x17c80010

Device      Boot  Start      End Sectors Size Id Type
/dev/sda1                2048 4196351 4194304  2G 83 Linux

Command (m for help): w
The partition table has been altered.
Syncing disks.
```

- Apply changes

```
22:40:14 | root@localhost | ~ | cmd:207 | cpu: 0.08 | mem:28% | disk:
| git: | ssh: | time:0s
🚀 partprobe
```